

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain

- a) Role of polymer in energy conservation and protection of environment
- b) Biodegradable polymers

Q.24 Explain manufacturing of vinyl chloride (Monomer) their properties and application.

Q.25 What is the need of polymer recycling? Explain primary recycling, secondary recycling, tertiary recycling.

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Roll No.

2nd Sem/ Plastic Technology Subject : Polymer Chemistry

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 PE is prepared from:

- a) Ethylene
- b) Vinyl Chloride
- c) Styrene
- d) Phenol

Q.2 The functional group of aldehyde is:

- a) -COOH
- b) -OH
- c) -CHO
- d) -NH₂

Q.3 Mechanical recycling is known as-

- a) Primary recycling method
- b) Secondary recycling method
- c) tertiary recycling method
- d) None of above

Q.4 Which force is strongest among secondary bonding forces?

- a) Dispersion force
- b) Dipole force
- c) Hydrogen bonding
- d) Van der Waals force

Q.5 Which polymer is biodegradable?

- a) PVC b) PE
- c) Nylon d) PLA

Q.6 Amorphous polymers have backbone chain arranged in _____.

- a) Ordered way b) Disordered way
- c) Both a and b d) NOT

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Write one functional group present in Base.

Q.8 Expand the term IUPAC.

Q.9 Define term Monomer.

Q.10 Write the general formula of alkyne.

Q.11 Write example of one crystalline polymer.

Q.12 Define Organic compound.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Write IUPAC Names of

(a) $\text{CH}_3\text{-CH}_2\text{-CH=CH}_2$ (b) $\text{CH}_3\text{-CH}_2\text{-CHO}$

Q.14 Define the term viscosity. How the viscosity of dilute polymer solution is measured?

Q.15 With one example define cis and trans isomerism.

Q.16 Explain isomerism and its types.

Q.17 Define poly functional compounds with example.

Q.18 Explain source of monomer from petrochemical.

Q.19 What is degraded polymer? Write two factors / agents which affect polymer degradation.

Q.20 Write chemical structure of (a) methanol and
(b) propanal.

Q.21 Define macromolecules.

Q.22 Explain Primary and secondary recycling of polymers.